

Electronically Overhauling the *Homo Sapien*



Dilin Anand is a senior assistant editor at EFM

Neil Harbisson, 32, can receive phone calls directly into his head. He can perceive colours invisible to the human eye. He has an antenna that sprouts from the occipital bone in his Bluetooth-enabled skull. Neil is not a sci-fi movie character, nor from the future; he is the world's first legally-recognised cyborg.

Just like Neil, wearing and later embedding electronics is now becoming part of life for an increasing number of people. "Wearables, a nicer name for wearable computers, are not just technology strapped onto body parts. These will play a significant part in the Internet of Things (IoT), which aims to connect everyone and everything with each other," says Deepak Thomas, India sales manager at CSR.

While some people wear electronics to help them perceive their surroundings more deeply, others wear these to keep track of calories, distance walked, heart-condition and other parameters.

Enabling smarter humans

Did you know there are people who can sense their laptops beginning to stall, or if their hard drives have begun to spin-up under load? An article by Quinn Norton from Wired had talked about this almost a decade back. Fast-forward to the present, and adding new sensor technology that complements or

supplements our existing five senses has already gained steam.

The Internet went wild with news coverage when people began adding rare Earth magnets implanted in their fingertips, but in 2015, you have dedicated bio-hacker communities, like biohack.me, set-up to support grinders (people who practice body modification to extend and improve human capacities). With this trend already up and running, humanity just might move at least a few steps closer to becoming entities like the cybernetic *Borg* species from the fictional world of Star Trek.

What new role does sensing play in this

After measuring calories, steps and distance, measuring heart rate is the new goal every fitness device developer aims to perfect this year. Vendors like Valencell specialise in clinically-validated biometric technology, and have come out with newer heart-rate-monitoring technology, powered by better sensors and algorithms. Their PerformTek sensor technology is licensed by the likes of Jabra, Scosche, LG and ATLAS. One important feature of their photo-detector-powered heart-rate-monitoring technology is its ability to be scalable to multiple form factors, like earbuds, armbands and wrist devices. Companies such as Belgium based imec are working on optical sensors that can give accurate pulse and blood oxygen levels, and wearable ECG sensors to monitor your heart function.

Perhaps, the most interesting of all is Xelflex technology, which enables a garment itself to become the sensor. Developed by Cambridge Consultants, the fibre-optic-thread based garment provides the user with guidance and feedback on his or her movements and can be used for a variety of purposes, like sports and fitness activities. A

